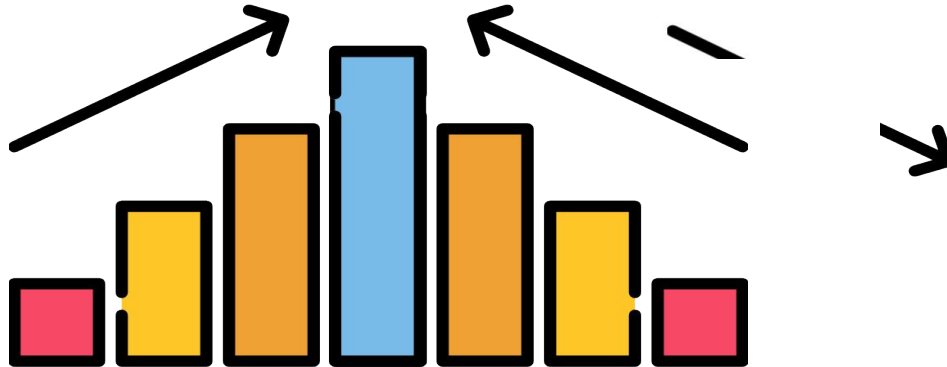


Monotonicity



Monotonic stack

- A stack whose elements are **monotonically** increasing or decreasing.
- Useful when we're looking for the next larger/smaller element.
- For a mono-decreasing stack:
 - we need to pop smaller elements before pushing.
 - By continually removing smaller elements and retaining greater ones, the mono-decreasing stack refines the result to achieve the maximum lexicographical greatness possible.



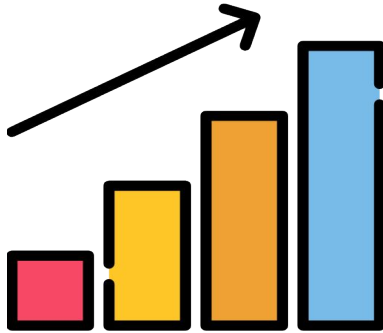
Practice

[Problem](#)

Practice

[Problem](#)

Monotonic Stack Application



- It gives you how far a value spans as a **maximum** or **minimum** in the given array.

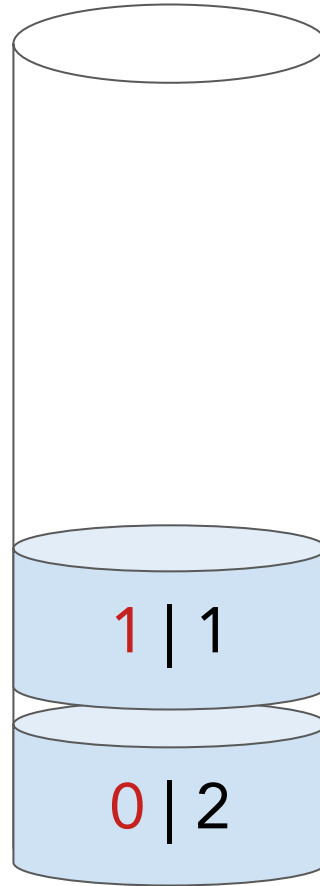
[2, 1, 3, 5, 6, 1]
0 1 2 3 4 5

Monotonically decreasing
stack



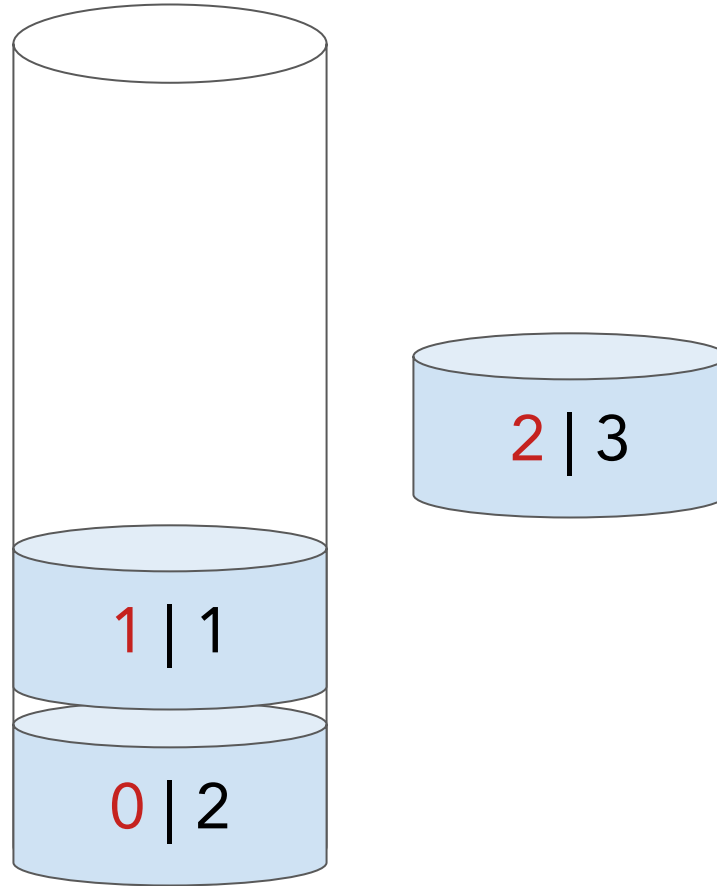
[2, 1, 3, 5, 6, 1]
0 1 2 3 4 5

Monotonically decreasing
stack



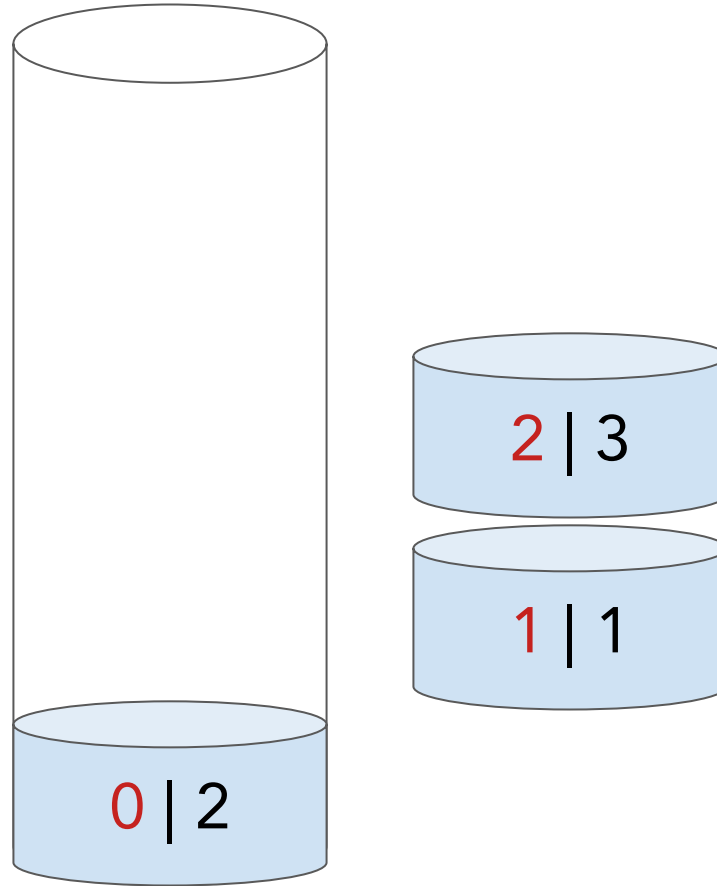
[2, 1, 3, 5, 6, 1]
0 1 2 3 4 5

Monotonically decreasing
stack



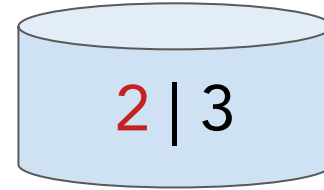
[2, 1, 3, 5, 6, 1]
0 1 2 3 4 5

Monotonically decreasing
stack



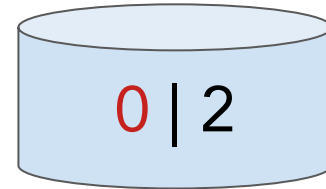
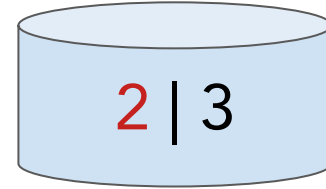
[2, 1, 3, 5, 6, 1]
0 1 2 3 4 5

Monotonically decreasing
stack



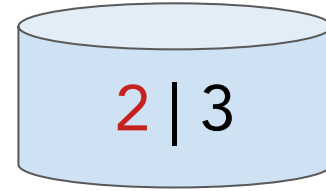
[2, 1, 3, 5, 6, 1]
0 1 2 3 4 5

Monotonically decreasing
stack



[2, 1, 3, 5, 6, 1]
0 1 2 3 4 5

Monotonically decreasing
stack



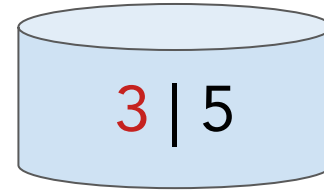
[2, 1, 3, 5, 6, 1]
0 1 2 3 4 5

Monotonically decreasing
stack



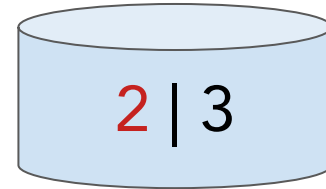
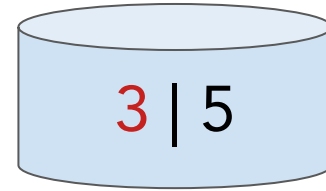
[2, 1, 3, 5, 6, 1]
0 1 2 3 4 5

Monotonically decreasing
stack



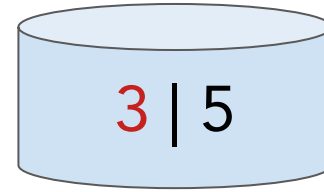
[2, 1, 3, 5, 6, 1]
0 1 2 3 4 5

Monotonically decreasing
stack



[2, 1, 3, 5, 6, 1]
0 1 2 3 4 5

Monotonically decreasing
stack



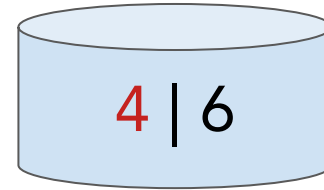
[2, 1, 3, 5, 6, 1]
0 1 2 3 4 5

Monotonically decreasing
stack



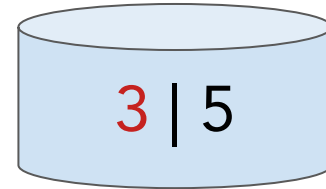
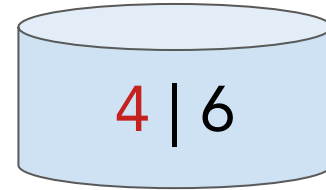
[2, 1, 3, 5, 6, 1]
0 1 2 3 4 5

Monotonically decreasing
stack



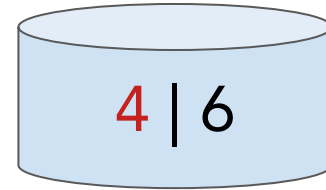
[2, 1, 3, 5, 6, 1]
0 1 2 3 4 5

Monotonically decreasing
stack



[2, 1, 3, 5, 6, 1]
0 1 2 3 4 5

Monotonically decreasing
stack



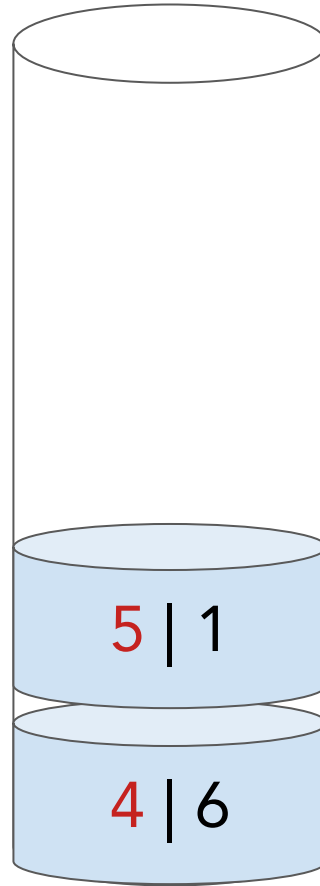
[2, 1, 3, 5, 6, 1]
0 1 2 3 4 5

Monotonically decreasing
stack



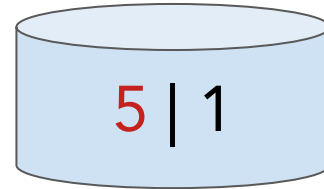
[2, 1, 3, 5, 6, 1]
0 1 2 3 4 5

Monotonically decreasing
stack



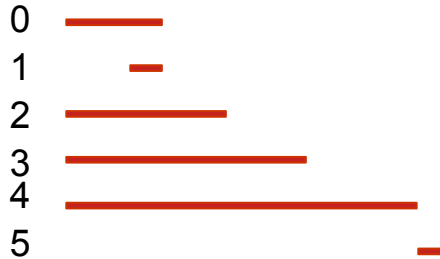
[2, 1, 3, 5, 6, 1]
0 1 2 3 4 5

Monotonically decreasing
stack



[2, 1, 3, 5, 6, 1]

0 1 2 3 4 5



Index Spans:

1: $1 \leq \dots < 2$

0: $0 \leq \dots < 2$

2: $0 \leq \dots < 3$

3: $0 \leq \dots < 4$

5: $4 \leq \dots < 5$

Monotonic Queue

- A queue whose elements are monotonically increasing or decreasing.
- For a mono-decreasing Queue:
 - To push an element e , the process begins with the rear element, where elements smaller than e are sequentially popped out.

Pair Programming

Problem

Time and Space Complexity

- The time and space complexity for monotonic stack and queue operations are the same as stack and queue operations.
 - **Why?**

Check point

Pitfalls & Opportunities

- Be careful of how to handle equality
 - Should we pop elements in the monotonic stack/queue that are equal?
- Check if stack/queue is empty before accessing/removing
- For greater problems, usually use a monotonically increasing stack
- For smaller problems, usually use a monotonically decreasing stack

Practice Questions

Stacks

- [Valid Parentheses](#)
- [Simplify Path](#)
- [Evaluate Reverse Polish Notation](#)
- [Score of parenthesis](#)
- [Backspace String Compare](#)

Queues

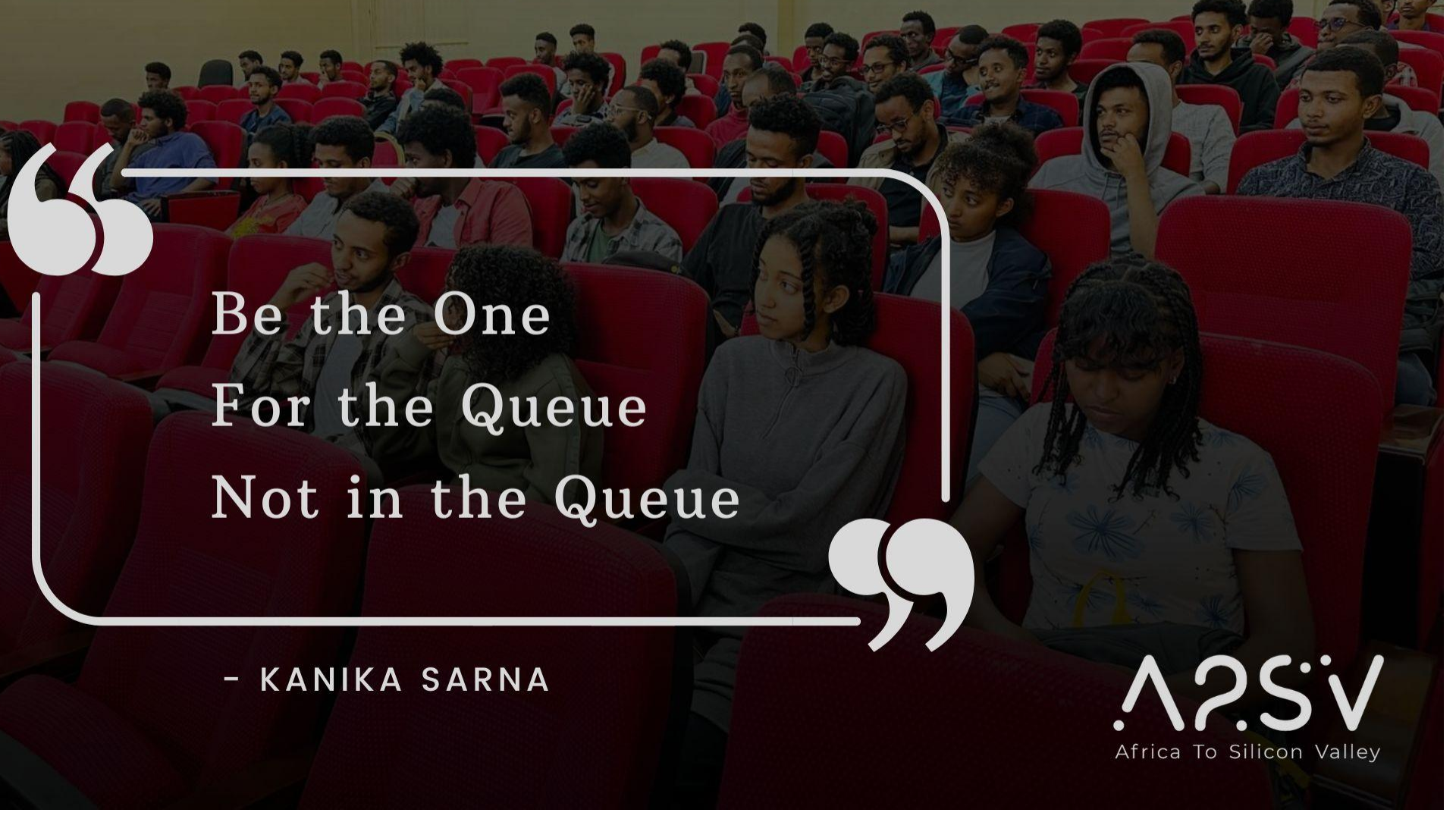
- [Number of recent calls](#)
- [Find consecutive integers](#)
- [Design Circular Deque](#)
- [Implement Queue using Stack](#)
- [Shortest subarray with sum at least K](#)

Monotonic

- [Car Fleet](#)
- [Remove duplicates](#)
- [Sum of subarray minimum](#)
- [Remove k digits](#)
- [132 Pattern](#)

Resources

[A comprehensive guide and template for monotonic stack based problems](#)



“
Be the One
For the Queue
Not in the Queue
”

– KANIKA SARNA

A2SV
Africa To Silicon Valley